

Assignment 7 - C++ - strings, containers and classes

100%

28.11.2023

Poengsum frakoblet modus:

100%



Legg til kommentar

Anonym vurdering: **nei**

15.12.2023

▼ Detaljer

Introduction

This assignment covers the C++ topics at the end of the course. Since this single assignment covers multiple lectures you might want to wait until the relevant lecture has been held. The two first tasks are similar to tasks you've already done with C, so take inspiration from those but consider how this could be solved more easily with C++ and the C++ Standard Library.

Note: You are not allowed to use any C Standard Library functions for these tasks, meaning no `string.h`, `stdio.h`, `stdlib.h` and so on.

Note 2: Choose C++ *Executable* and C++ version 14 when creating the projects for this assignment.

7.1

Directory name: `ikt101g23h/assignments/solutions/assignment_7_1`

Relevant lecture: C++ IO, strings

Write an application that asks for two strings and:

- Tells if the strings are equal or not
- Tells if one string is a substring of the other

Areas of interest: `std::cin`, `std::cout`, `std::string`

Expected output:

The strings are equal / The strings are not equal

First string is substring of last string / Last string is substring of first string

There are no substrings

7.2

Directory name: ikt101g23h/assignments/solutions/assignment_7_2

Relevant lecture: C++ containers

Write an application that asks for numbers until 0 is given as input and:

- Prints out the average
- Prints out the median
- Prints out the numbers sorted, in descending order

Input: 1 2 3 4 5 6 7 8 9 10 0 // Numbers are just an example

Expected output:

Average : 5.5

Median : 5.5

Descending : 10 9 8 7 6 5 4 3 2 1

Note: The 0 should not be counted or used in the calculations. And no more than 4 decimals.

Areas of interest: std::cin, std::list or std::vector, <algorithm> header

7.3

Directory name: ikt101g23h/assignments/solutions/assignment_7_3

Relevant lecture: C++ classes

Write an application that:

1. Asks for the name of a student
2. asks for the name of a course and the grade the student got
3. Does this until course name is "stop"
4. Then asks for a new student until the name is "stop"
5. Then prints out the grades gathered for each student

Areas of interest: C++ classes, std::getline, std::map or std::vector or std::list

Expected output: Lars - ikt101 - C (*for each student*)

Test locally

see *Local testing with Docker* under Assignments.

Test the solution on your own computer. You can fix your code and retest as many times as you want.

Deliver and test in Bamboo

See *Delivery and Bamboo testing* under Assignments.

Finally deliver and test the solution in Bamboo. This is what updates your score in Canvas which is the score I can see. You can test as many times as you want here as well up until the final deadline.